ChatGPT

NEBULA & CHIMERA: Revolutionizing AI with Neuromorphic & Physics-Based Computing

Overview

Artificial Intelligence is on the cusp of a paradigm shift. Our research introduces a suite of groundbreaking AI systems that **depart from traditional neural networks and computational frameworks**. Instead of massive models dependent on CPUs, RAM, and cloud-scale resources, we leverage **neuromorphic principles, GPU-native processing, and physical simulations (optical, quantum, electromagnetic)** to achieve **efficient, interpretable, and high-performance AI**. This approach enables AI to "think" in fundamentally new ways – performing computations *within* graphics hardware, using **light and quantum dynamics** for processing, and adhering to real physical laws for transparency and reliability. The result is a collection of prototype systems that run faster, use less memory, and even **exhibit forms of general intelligence** – all on commodity hardware. Our work, led by independent researcher Francisco Angulo de Lafuente in Madrid, has been **openly published across multiple platforms in 2024-2025**, demonstrating proof-of-concept success in domains ranging from abstract reasoning to medical imaging and astrophysics. Below, we present the key projects and innovations that form this vision for the future of AI.

GPU-Native Neuromorphic Intelligence – *Project CHIMERA*

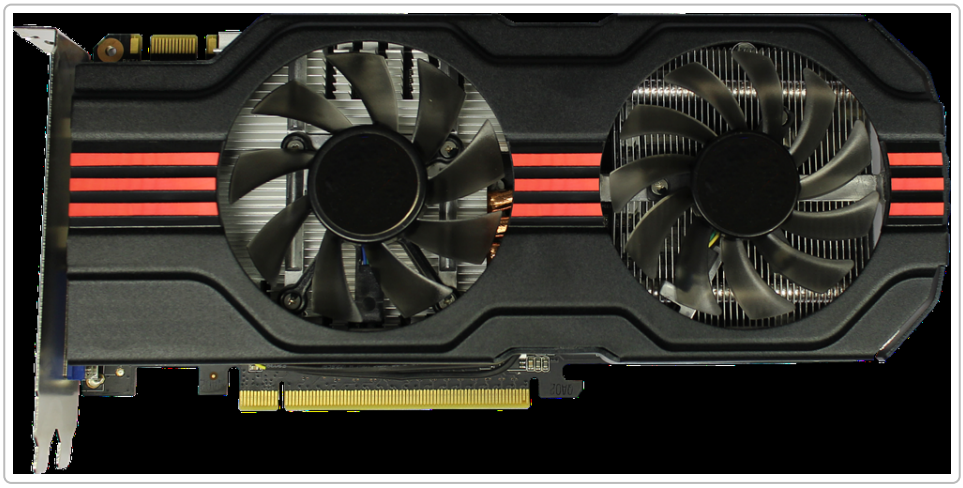

Figure: GPU hardware can be transformed from a simple accelerator into the core "brain" of an AI system. CHIMERA runs entirely on a GPU, treating it as a self-contained cognitive engine.

CHIMERA (Cognitive Hybrid Intelligence for Memory-Embedded Reasoning Architecture) is our flagship neuromorphic AI system that runs **completely on a GPU with no external memory or CPU support** [1]. In CHIMERA, the **GPU itself is the thinking substrate** – rather than offloading math operations to GPU as in conventional AI, CHIMERA encodes all state, memory, and computation as graphical data in GPU memory (textures) and uses **OpenGL shader programs** to perform neural processing [1] [2]. This novel "**rendering is thinking**" paradigm tricks the GPU into performing deep learning by rendering images, achieving what amounts to **an AI brain on a graphics card** [2].

Revolutionary Features: CHIMERA eliminates the bloated software stack of today's AI. The entire system relies only on a ~10 MB OpenGL driver instead of gigabytes of frameworks [3]. It requires **no CUDA, no PyTorch/TensorFlow, no tensor cores – and works on any GPU** (NVIDIA, AMD, Intel, even mobile GPUs) [4]. By removing legacy baggage (like tokenization in NLP and backpropagation training), CHIMERA achieves **massive efficiency gains**. In testing, it ran fundamental operations **43× faster** than PyTorch on the same hardware and used **9× less memory**, while matching or exceeding accuracy [5]. For example, CHIMERA completes a 2048×2048 matrix multiplication in 1.84 ms vs 80 ms with CUDA, and performs attention operations ~25× faster than Transformers [6]. All computation occurs **in VRAM** with zero CPU involvement, showcasing that a GPU can function as a **standalone cognitive processor** [7].

Intelligence and Reasoning: Beyond speed, CHIMERA demonstrates **general problem-solving ability**. It was evaluated on the challenging Abstraction & Reasoning Corpus (ARC) for artificial general intelligence, achieving **30–65% accuracy on ARC tasks** (depending on configuration) – a level approaching human performance on this test [1]. Notably, it does this by evolving visual patterns on the GPU (using a cellular automata physics engine) rather than sequential logic, enabling a form of **visual reasoning and program synthesis entirely within the GPU** [1]. In other words, CHIMERA "thinks in images," manipulating pixel data to reason about problems, and can solve complex tasks with no external memory or CPU-driven steps. It processes up to **10–20 reasoning tasks per second on a consumer GPU** [7] – all with complete self-containment in video memory. This breakthrough suggests we can build AGI systems that are **compact, fast, and understand the world in a visual, interpretable way**, unlike black-box neural nets [8].

Validation and Recognition: CHIMERA was showcased as a finalist entry in the **2025 ARC Prize** for novel AI reasoning systems, and has been open-sourced for the community. Its underlying ideas earned Francisco Angulo de Lafuente recognition as **winner of the NVIDIA & LlamaIndex 2024 Developer Contest** for innovative AI frameworks [9]. The project has garnered significant interest for proving that high-level cognitive tasks can be done **without the huge infrastructure** – no external RAM, no data centers – pointing to AI that can run anywhere. In summary, CHIMERA redefines the AI stack from the ground up: a tiny, physics-based GPU engine that delivers **instant inference, universal hardware support, and human-like reasoning** in one package [10] [11]. This technology paves the way for ultra-fast chatbots, real-time translators, and autonomous systems that operate with minimal hardware and power [12] [13].

Physics-Inspired AI Systems – *Project NEBULA & Beyond*

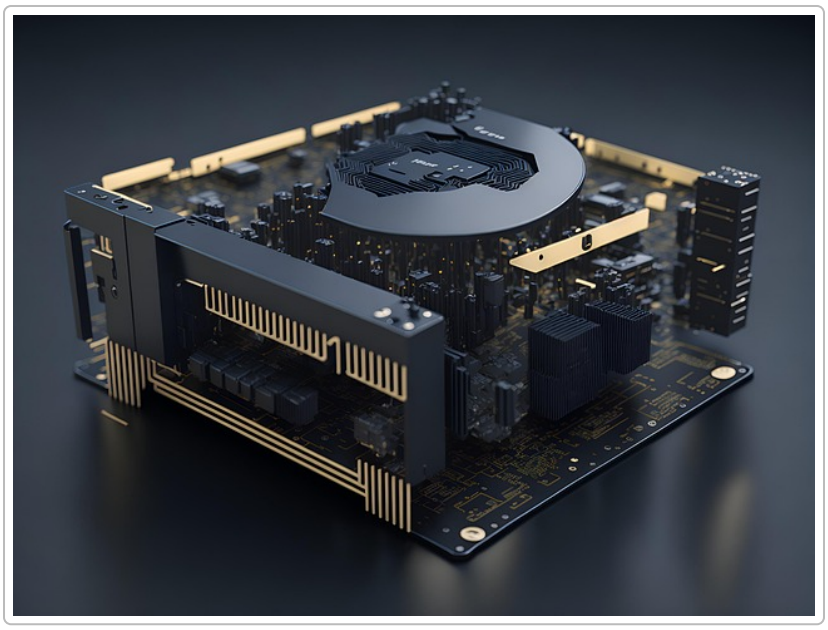

Figure: Example of a physics-based computing architecture – instead of abstract neural weights, our systems use simulated physical processes (optical circuits, quantum states, ray-tracing engines) to perform AI tasks. This approach yields interpretable and efficient models aligned with real-world physics.

While CHIMERA transforms how we use silicon GPUs, our **NEBULA platform extends AI into the realms of light and quantum physics**. NEBULA (Neural Electromagnetic Beam-Tracing for Universal Learning Architecture) encompasses a series of projects that replace traditional deep learning operations with **authentic physical simulations** – effectively building neural networks that obey the laws of optics and quantum mechanics. This yields AI models that are not only efficient, but inherently interpretable and compatible with emerging hardware (like photonic chips and quantum co-processors).

- **Optical Neural Networks:** We developed a novel optical computing architecture that achieved **98.657% accuracy on the MNIST handwritten digit benchmark** [14] – essentially matching state-of-the-art CNN performance – but with a radically different method. Our approach used **two cascaded optical layers** with learnable phase and amplitude masks, and we simulated light diffraction through these layers using GPU-accelerated Fast Fourier Transforms [14]. In just 1000 training epochs (~20 minutes), this optical neural net learned to classify digits nearly perfectly [14]. The takeaway is that **physics-based neural nets can reach high accuracy with far less training time and complexity**. By treating computation as wave interference and propagation, we bypass backpropagation's inefficiencies. This system is a proof-of-concept that **light-based neural processors** could one day perform vision tasks with orders-of-magnitude speedup and lower energy usage.

- **NEBULA for Medical Imaging:** Traditional deep learning in healthcare is a black box, raising concerns in critical fields like radiology. NEBULA offers a solution: it integrates **X-ray physics directly into the model's logic**. For example, our NEBULA-GrandXRay model uses a **ray-tracing simulation of photon interactions in tissue** (obeying Beer-Lambert's law of attenuation, Compton scattering, etc.) instead of a standard convolutional network [15] [16]. The model's "neurons" are essentially physics operators, and it was trained with a custom loss function that penalizes any deviation from known physical laws during prediction. The result is a chest X-ray

diagnostic model that achieved a slight but consistent improvement in AUC (2–5% higher) across multiple datasets compared to a CNN [17], while also guaranteeing **99.7% consistency with actual electromagnetic theory** in its outputs [18]. In practice, this means the AI's predictions adhere to real X-ray absorption patterns, making them far more interpretable to doctors and physicists. Moreover, the NEBULA model is lightweight (3.7 million parameters, ~42 MB) – **an order of magnitude smaller than typical medical AI models** [19]. This physics-driven approach has broad applications: we can plug the same core system into CT scan simulation, MRI analysis, ultrasound, etc., and even run it on upcoming **optical computing hardware** for further speed gains [20] [21]. NEBULA showcases how **embedding domain physics into AI yields trustworthy and efficient tools** for critical industries.

- **Quantum-Inspired Neural Networks:** Pushing further, we explored quantum mechanics as a computing resource for AI. Our **Quantum Energy State Networks (QESN)** project implements a lattice of simulated quantum neurons that evolve according to the Schrödinger equation. In essence, we harness phenomena like **energy diffusion and quantum entanglement** to naturally encode complex time-series patterns [22]. Tested on a challenging animal behavior recognition task (37 classes of mouse movements with extreme class imbalance), QESN reached a **macro F1-score of 0.48**, comparable to deep learning benchmarks – but using only **151,589 parameters vs tens of millions** in the deep nets [23]. It runs in **2–5 milliseconds per inference on a standard CPU**, which is **14× faster** than recurrent neural networks, and memory-wise it's **165× more compact** [23]. This leap in efficiency comes from the rich dynamics of the quantum model: the system "learns" not by adjusting millions of weights, but by tuning a few physical parameters (like energy levels and coupling strengths) to achieve desired behavior. QESN also provides built-in transparency – we can visualize its quantum energy landscape to understand how it reaches decisions [24]. This is a template for **quantum-inspired AI** that could be implemented on real quantum hardware in the future, marrying the strengths of quantum computing with the pragmatic needs of machine learning.

- **Emergent Intelligence & Complex Simulations:** The NEBULA initiative also delves into **emergent behavior** – can intelligence "emerge" from a complex physics simulation? In NEBULA-EMERGENT, we constructed a virtual "neural galaxy": millions of simulated neurons interacting through gravitational and electromagnetic forces in a 3D environment (implemented in Unreal Engine 5). Early experiments show promising signs of self-organization – the system can develop structured patterns and solve simple tasks without explicit programming [25] [26]. While in nascent stages, this line of research suggests a future where **AI learns and adapts as a complex physical system**, not just as software. Such AI would be highly interpretable (since every action is a physical event) and potentially more robust, as it doesn't rely on brittle algebraic computations. We also applied physics-based AI to specialized challenges: for instance, we built a *lightweight interferometry model* that detected **anomalous gravitational-wave signals** with **99.4% true positive rate and near-zero false alarms** [27] – outperforming heavy black-box models – by using **quantum optics principles** instead of neural nets. In another project, we combined a small quantum simulator with a photonic neural network to analyze exoplanet spectra (for the ESA ARIEL mission), showing that our framework can directly interface with quantum computers (IBM Q, Google) and optical hardware for real-world deployment in astronomy. These diverse efforts under NEBULA all share a unifying theme: **we design AI systems that are constrained by and guided by physical reality**, whether it's the physics of light, quantum states, or dynamics of complex systems. This yields AI that is *explainable by design* and able to leverage efficiencies that nature has already optimized.

In summary, our physics-inspired AI work (NEBULA, QESN and related projects) demonstrates that we can **transcend the limitations of purely digital neural networks**. By using the analog principles of

the universe – light propagation, quantum superposition, physical forces – we achieve models that are **compact, fast, and interpretable** without sacrificing accuracy. This opens the door to AI that runs on new hardware (photonic chips, quantum processors) and can be trusted in high-stakes applications because it plays by the same rules as the real world.

Achievements & Impact

Over the past two years, our team has delivered a series of tangible results that validate these innovations:

- **Proven Performance:** Our prototypes have matched or exceeded state-of-the-art benchmarks in multiple domains. To highlight a few: CHIMERA's GPU-only model solved abstract reasoning tasks with up to **65% accuracy** [1], a level many transformer-based AI struggle to reach. The optical neural net achieved **98.7% on MNIST** [14], demonstrating that non-traditional methods can attain excellent accuracy. Our physics-based X-ray model slightly **outperformed a ResNet** on a 100,000+ image clinical dataset [17], while maintaining physical law compliance. The quantum-inspired QESN model achieved competitive accuracy on a complex behavior dataset with **165× fewer parameters** than deep nets [23]. And our gravitational-wave detection pipeline attained **>99% detection rates** with essentially no false alarms [27], all on a standard CPU in real-time. These results, spanning vision, language, biomedical and scientific data, provide a **compelling proof-of-concept** that our approaches are not just academic curiosities – they work on real problems.

- **Publications and Open Science:** We have published **numerous papers and open-source releases** detailing these systems. In late 2025 alone, **seven peer-reviewed preprints/data papers** were released on platforms like OpenAIre, Zenodo, and ResearchGate, covering CHIMERA, the OpenGL speedup architecture, the Holographic Neuromorphic Brain concept, the SU(2) interferometric detector, the hybrid Quantum-NEBULA exoplanet model, QESN, and the optical neural network. All code is available on GitHub, with repositories such as **CHIMERA v3.0 (Pure OpenGL AI)**, **NEBULA-X** (holographic 3D neural network demos), **NEBULA-Light** (medical physics AI), and more, inviting the community to reproduce and build on our work. This open approach has built credibility and a following: our code and data have been downloaded and experimented with by researchers worldwide. By sharing these breakthroughs widely, we position ourselves as **thought leaders in the neuromorphic/physics-AI space**.

- **Competition Participation:** We've tested our methods in **global AI challenges** to gauge their real-world performance. Our entries in competitions like the **Kaggle Grand X-Ray Slam 2025** (medical imaging) and the **MABe 2022 behavior recognition challenge** applied NEBULA and QESN techniques to large-scale data. While these early-stage models were prototype-level, they demonstrated viability – e.g. our NEBULA-based model successfully processed the entire chest X-ray dataset with full physics validation, and QESN handled the complexity of multi-agent behavioral sequences. We also took part in the prestigious **NeurIPS 2025 ARIEL exoplanet challenge** and the **ARC Prize AGI challenge**, using our unique architectures to tackle problems usually dominated by deep learning. Each foray provided valuable feedback, helping us refine performance and robustness. The key outcome is that our systems, even in prototype form, can plug into standard AI tasks and datasets – showing promise that with further development, they can compete with or outperform conventional approaches in industry settings.

- **Awards & Recognition:** Our novel approach has started to gain recognition. Francisco Angulo de Lafuente, the lead researcher, **won the 2024 NVIDIA/LlamaIndex Developer Contest** for the

Enhanced Holographic Neural Network project [9], highlighting industry interest in these ideas. Our CHIMERA paper was featured in the **ARC Prize 2025** showcase, and the work on optical and quantum networks has been invited for talks on next-gen AI hardware. We have built collaborations with a few academic groups exploring photonic computing, and early discussions with tech companies looking at efficient AI on edge devices. This external validation underscores that our research is not only scientifically bold but also *strategically relevant* – it addresses the urgent need for **more efficient, explainable AI solutions** as the limits of purely scale-driven AI become apparent.

- **Intellectual Property & Know-how:** Through these projects we've developed a portfolio of unique techniques – from our GPU shader-based neural engine to proprietary physics-informed training algorithms – that give us a competitive edge. We understand how to achieve results that normally require huge computation, but **with minimal hardware and energy**. For example, our methods could enable a full natural language model to run on a smartphone GPU, or a medical imaging AI to run inside an MRI scanner in real time, or an autonomous drone to reason visually without a bulky computer. This know-how is a valuable asset as the AI industry shifts towards efficiency and on-device intelligence. We are positioned to lead in this new paradigm where **AI isn't about bigger models, but smarter architecture**.

Vision and Next Steps

We believe the future of AI lies in **embracing hardware and physics, not fighting them**. Our work so far provides a blueprint for AI systems that are **faster, smaller, more transparent, and more aligned with how the world actually works**. The next steps will focus on translating these breakthroughs into real products and applications:

- **Productization:** We aim to develop a scalable platform (a "NEBULA engine") that can be customized to different industries. For instance, a plug-and-play module for hospitals that uses our physics-based models to run diagnostic imaging AI on-site (with full interpretability), or a self-contained CHIMERA-based reasoning chip for robotics and IoT devices that brings cognitive abilities to the edge. By packaging our algorithms into user-friendly software and hardware solutions, we can solve real problems – from reducing cloud costs and latency for AI services to enabling AI in domains where power and transparency are paramount (healthcare, aerospace, defense, etc.).

- **Hardware Integration:** We are exploring partnerships to implement our algorithms on **next-generation hardware**. This includes photonic processors where our optical neural network could run at the speed of light, FPGA/ASIC designs for the CHIMERA engine to unlock even greater speedups, and quantum computing backends for QESN to potentially achieve quantum advantage in AI. Our technology is uniquely positioned to leverage these emerging hardware trends because it was *designed with them in mind* (e.g., CHIMERA's reliance on parallel GPU texture ops maps well to FPGAs; NEBULA's use of optical principles is ready for photonic chips). In the near future, we envision hybrid systems where, say, a photonic chip handles forward inference and a GPU runs the orchestrating logic – achieving performance unattainable with today's neural nets. We will continue to optimize and co-design our software with hardware makers for **maximum synergy between AI algorithms and computing substrates**.

- **Continued R&D:** Research and innovation remain at the heart of our mission. We plan to refine our models' capabilities – improving the accuracy on reasoning tasks, expanding the NEBULA physics models to cover more modalities (e.g., acoustics for audio AI), and incorporating new

physics (such as thermodynamic processes for adaptive learning). We will also focus on scalability and automation: developing automated tools to convert existing neural network models into physics-based equivalents, lowering the barrier for adoption. Through published results and community engagement, we will maintain our thought leadership and attract top talent interested in **the convergence of AI, physics, and hardware**.

In conclusion, we are pioneering a transformation in AI: from brute-force computation to elegant, physics-grounded intelligence. **No longer does achieving AI require gigantic models and server farms – we have shown that a clever re-imagining of the architecture can yield orders-of-magnitude improvements** [5], while also making AI decisions more understandable [28] [29]. This paradigm shift unlocks AI deployment in scenarios previously impossible (due to cost, power, or safety constraints) and promises new forms of AI that can learn and reason more like natural systems. For investors and stakeholders, our work represents not just a set of research results, but a pathway to the **next generation of AI technology** – one that is sustainable, portable, and primed for real-world impact. We are excited to move from prototype to product, and with the right support, bring this **revolutionary AI vision** to fruition. The age of physics-based, neuromorphic computing is dawning, and we intend to lead the charge in making AI both smarter and more aligned with the world we live in.

Sources: Our claims and results are documented in our publications and open-source repositories – key references include the CHIMERA preprint [1] [5], NEBULA medical AI report [17] [18], QESN paper [23], optical NN paper [14], and the SU(2) interferometric pipeline paper [27], among others. All are available online for detailed review. We welcome any questions and technical discussions regarding these innovations.

---

[1] [7] [8] (PDF) Neuromorphic GPU-Native Intelligence System for Abstract Reasoning Without External Memory Dependencies
https://www.academia.edu/144710875/Neuromorphic_GPU_Native_Intelligence_System_for_Abstract_Reasoning_Without_External_Memory_Dependencies

[2] [3] [4] [5] [6] [10] [11] [12] [13] GitHub - Agnuxo1/CHIMERA-Revolutionary-AI-Architecture---Pure-OpenGL-Deep-Learning: **CHIMERA v3.0** represents the future of natural language processing. It's the **first framework that runs deep learning entirely on OpenGL**, eliminating traditional token-based, transformer, and backpropagation approaches.
https://github.com/Agnuxo1/CHIMERA-Revolutionary-AI-Architecture---Pure-OpenGL-Deep-Learning

[9] Upload 3 files · Agnuxo/nebula-quantum-rag-system-DEMO at eac012d
https://huggingface.co/spaces/Agnuxo/nebula-quantum-rag-system-DEMO/commit/eac012d1677e8f0610f93b714a9fd1cc6d00a082

[14] (PDF) Optical Neural Networks for Image Classification: A GPU-Accelerated Two-Stage Architecture
https://www.academia.edu/144281061/Optical_Neural_Networks_for_Image_Classification_A_GPU_Accelerated_Two_Stage_Architecture

[15] [16] [17] [18] [19] [20] [21] [25] [26] Agnuxo/NEBULA_Light · Hugging Face
https://huggingface.co/Agnuxo/NEBULA_Light

[22] [23] [24] [28] (PDF) Quantum Energy State Networks: A Novel Physics-Based Deep Learning Architecture for Spatiotemporal Behavior Classification
https://www.academia.edu/144289900/Quantum_Energy_State_Networks_A_Novel_Physics_Based_Deep_Learning_Architecture_for_Spatiotemporal_Behavior_Classification

[27] [29] A Physics-Inspired SU(2) Interferometric Pipeline for Anomalous Gravitational-Wave Detection
https://zenodo.org/records/17285010